

Variable shadowing in Java occurs when a variable declared within a specific inner scope (like a method, constructor, or block) has the exact same name as a variable declared in an outer scope (like a class field). When this happens, the inner variable **shadows** (masks) the outer variable, making the outer variable temporarily inaccessible by its simple name within that specific inner block.

Code:

```
public class ShadowingExample {  
    // Outer scope: Instance variable  
    String name = "Global Reference";  
  
    public void display() {  
        // Inner scope: Local variable shadows the instance variable  
        String name = "Local Reference";  
  
        // Prints the local variable because it takes precedence  
        System.out.println(name);  
  
        // Accessing the shadowed instance variable using 'this'  
        System.out.println(this.name);  
    }  
  
    public static void main(String[] args) {  
        ShadowingExample obj = new ShadowingExample();  
        obj.display();  
    }  
}
```

Output:

Local Reference
Global Reference

Key Rules of Shadowing in Java

- **Scope Precedence:** Java always prioritizes the variable closest to the current execution block.
- **No Redefinition Errors:** Declaring a local variable with the same name as a class instance variable is completely valid and will not trigger a compiler error.
- **Same-Scope Restriction:** You cannot shadow a local variable within the exact same method block or loop block; trying to redeclare *int x* twice inside the same method body will cause a compile-time error.
- **Data Types:** Shadowing occurs even if the inner variable and outer variable have completely different data types.

How to Access Shadowed Variables

When a variable is hidden by shadowing, you can still access it using explicit qualifiers:

- `this.variableName`: Used to access an instance variable from an inner instance method or constructor.
- `ClassName.variableName`: Used to access a static class variable if it is shadowed by a local variable